Physics 4240H: Assignment III

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Problem 1 (14.4)**.** Two linearly polarized beams are given by

$$\mathbf{E}_1 = E_{01}(\hat{\mathbf{x}} - \hat{\mathbf{y}})\cos(kz - \omega t) \quad \text{and}$$

$$\mathbf{E}_2 = E_{02}(\sqrt{3}\hat{\mathbf{x}} + \hat{\mathbf{y}})\cos(kz - \omega t).$$

Determine the angle between their directions of polarization by

(a) forming their Jones vectors and finding the orientation of each field

(b) forming the dot product of their vector amplitudes.

Solution 1.

(a) $\mathbf{E}_1$ has a π phase difference between the x and y components, so has Jones vector

$$\mathbf{V}_1 = E_{01}\begin{pmatrix}1\\ e^{i\pi}\end{pmatrix} = E_{01}\begin{pmatrix}1\\ -1\end{pmatrix}.$$

$\mathbf{E}_2$ has Jones vector

$$\mathbf{V}_2 = E_{02}\begin{pmatrix}\sqrt{3}\\ 1\end{pmatrix}.$$

These respectively have angles to the positive x-axis given by

$$\theta_1 = \arctan(-1) = -\frac{\pi}{4}$$

and

$$\theta_2 = \arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}.$$

The angle between these two is $5\pi/12$.

(b) We know that $\mathbf{a}\cdot\mathbf{b} = a_x b_x + a_y b_y = |\mathbf{a}||\mathbf{b}|\cos\theta$ where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$. For us this is

$$\sqrt{3} - 1 = \sqrt{8}\cos\theta \implies \theta = \arccos\left(\frac{\sqrt{3}-1}{\sqrt{8}}\right) = \frac{5\pi}{12},$$

W[5].

Problem 2 (14.9)**.** Specify the polarization state and field amplitude(s) represented by each of the following Jones vectors.

$$\text{(a) } \begin{pmatrix} 3i \\ i \end{pmatrix}\text{; (b) } \begin{pmatrix} i \\ 1 \end{pmatrix}\text{; (c) } \begin{pmatrix} 4i \\ 5 \end{pmatrix}\text{; (d) } \begin{pmatrix} 5 \\ 0 \end{pmatrix}\text{; (e) } \begin{pmatrix} 2 \\ 2i \end{pmatrix}\text{; (f) } \begin{pmatrix} 2 \\ 3 \end{pmatrix}\text{; (g) } \begin{pmatrix} 2 \\ 6+8i \end{pmatrix}.$$

Solution 2.

(a) Factoring out an i, the vector clearly represents linearly polarized light with an overall phase and amplitude $\sqrt{10}$ at angle 18.4° from the positive x-axis.

(b) Here we can factor out an i yielding $i(1, -i)$, right-circular polarization with an overall phase and amplitude 1.

(c) Again we factor out an i and obtain $i(4, -5i)$, right-elliptical polarization with an overall π phase and horizontal amplitude 4 and vertical amplitude 5, meaning the ellipse has a major axis lying along the vertical.

(d) Regular linearly polarized light along the horizontal with amplitude 5.

(e) Left circular polarization, amplitude 2.

(f) Linear polarization with amplitude $\sqrt{13}$ at an angle of 56.3°.

(g) Left-elliptical polarization, amplitudes 2 in the horizontal and 10 in the vertical, axis at an angle of $-7°$.

Problem 3 (14.13)**.** Linearly polarized light with a horizontal transmission axis is sent through a linear polarizer with its transmission axis at 45° and then through a $\lambda/4$-plate with its slow axis horizontal. Use the Jones matrix technique to determine and describe the transmitted light.

Solution 3. Light initially incident with jones vector

$$\mathbf{V} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

is subject to a linear polarizer at 45°, represented by

$$\underline{\underline{M}}_{45°} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

then to a $\lambda/4$ plate with SA horizontal,

$$\underline{\underline{M}}_{\lambda/4} = e^{-i\pi/4}\begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix}.$$

Chaining these then the outgoing light looks like

$$\mathbf{V}' = \underline{\underline{M}}_{\lambda/4}\underline{\underline{M}}_{45°}\mathbf{V} = \frac{e^{-i\pi/4}}{2}\begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{e^{-i\pi/4}}{2}\begin{pmatrix} 1 \\ -i \end{pmatrix}$$

which has a global phase given by $e^{i\pi/4}$ but also a π phase delay between the horizontal and vertical components due to the negative vertical component. The light is right-circularly polarized, with amplitude $1/4\sqrt{2}$.

Problem 4 (IT14)**.** A beam of monochromatic left-circularly polarized light traveling along the z-axis is incident on an eighth-wave plate (that is, a $\lambda/8$-plate), which may be rotated so its slow axis lies at any desired angle to the horizontal (this being the x-axis of our coordinate system).

(i) Which of the following expressions for the real, physical, electric field best describes the incident light?

(ii) If the $\lambda/8$-plate is oriented so that its slow axis is *horizontal* then which Jones matrix best represents the $\lambda/8$-plate with this orientation?

(iii) Which of the following expression then best represents the Jones vector, *and* the description of the polarization, of the light transmitted through the $\lambda/8$-plate when oriented as described in part (ii)?

(iv) If the $\lambda/8$-plate is rotated from the orientation of case (ii) by and angle θ clockwise when looking into the beam, then which Jones matrix best represents the $\lambda/8$-plate with this orientation? (It may be useful here to refer to footnote 8).

Solution 4.

(i) For left-circular polarization there needs to be a $\pi/2$ phase difference between the horizontal and vertical components, leaving (D) and (B). The answer must be (D) with phase $+\pi/2$ as left-circular polarization has Jones vector $(1, i)$ and $e^{i\pi/2} = i$.

(ii) We want to add a phase of $\pi/4$ to the horizontal axis, alternatively we retard the vertical component by $\pi/4$, meaning answer (A) must be correct.

(iii) We can apply the matrix in (ii) $(1, i)$ which yields a new Jones vector,

$$\mathbf{V}_{out} = \begin{pmatrix} 1 \\ e^{i\pi/4} \end{pmatrix} = \begin{pmatrix} 1 \\ \frac{1}{\sqrt{2}}(1+i) \end{pmatrix},$$

answer (B).

(iv) We apply the change of basis (in linear algebra terms) from the xy basis to the rotated one by

$$\begin{aligned} \underline{\underline{RM}}_{\lambda/8}\underline{\underline{R}}^T &= \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & e^{-i\pi/4} \end{pmatrix} \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \\ &= \begin{pmatrix} \cos^2(\theta) + e^{-i\pi/4}\sin^2(\theta) & (e^{-i\pi/4}-1)\cos(\theta)\sin(\theta) \\ (e^{-i\pi/4}-1)\sin(\theta)\cos(\theta) & \sin^2(\theta) + e^{-i\pi/4}\cos^2(\theta) \end{pmatrix} \end{aligned}$$

after some looking, answer (C).

Problem 5 (15.1)**.** Initially unpolarized light passes in turn through three linear polarizers with transmission axes at 0°, 30°, and 60° respectively, relative to the horizontal. What is the intensity of the emerging light, expressed as a percentage of the incident intensity?

Solution 5. We can directly apply Malus' law. The light sees an initial intensity decrease of 1/2 due to being unpolarized and forced to adopt one polarization state, then a decrease of $\cos^2(30°) = 3/4$ then another decrease of $\cos^2(30°) = 1/4$ because the light is now at an angle of 30° to the 60° polarizer. This gives a total decrease of $9/2 \cdot 4 \cdot 4 = 9/32$, 28.125% of the incident light.

Problem 6 (15.4)**.** How thick should a $\lambda/2$-plate made from cleaved mica be for an application where laser light of wavelength 632.8 nm is being used? Consider refractive indices of 1.599 and 1.594.

Solution 6. We want to have one retarded by 316.4 nm, meaning we need $\Delta\phi = \pi$ or

$$d = \frac{\Delta\phi\lambda_0}{2\pi\left|n_\perp - n_\parallel\right|}.$$

Taking $n_\perp = 1.599$ and $n_\parallel = 1.594$ we get

$$d \approx 60.3\,\mu\text{m}.$$

Problem 7 (15.12)**.** Light strikes a water surface at an angle such that the reflected light is completely linearly polarized.

(a) What is the angle of incidence?

(b) The light refracted into the water is then intercepted by the top smooth surface of a block of glass with an index of refraction of 1.50. The light reflected from the glass is completely linearly polarized. What is the angle between the glass and water surfaces?

Solution 7.

(a) For the reflection to be linearly polarized, the incident light arrived at Brewster's angle, $\theta_I = \arctan(1.33) \approx 53°$.

(b) Brewster's angle for water to glass is $\theta_B = \arctan(1.5/1.33) \approx 48°$. Snell's law tells us that, because the beam entered the water at $\theta_I = \arctan(1.33)$, θ_T, the angle at which the light travels through the water is $\arcsin(\sin(\arctan(1.33))/1.33) = \theta_T \approx 37°$ which is the angle at which the light strikes the glass, minus the angle at which the glass sits. We want $\theta_T + \theta_G$ to be Brewster's angle for water to glass (θ_G is the glass-normal angle, relative to the water) so

$$\theta_B - \theta_T = \theta_G \approx 11.5°.$$

Problem 8 (IT15)**.** A beam of white light, traveling along the z-axis, illuminates a sample of dilute molecules, as indicated in Figure 15.7. We shall treat the dashed ray shown in this figure as forming an angle of $\gamma_2 = 30°$ with the xz-plane, with its projection onto this plane lying midway between the x-axis and the z-axis (so that $\gamma_1 = 45°$). We wish to explore the scattered light for particular polarizations of the incident light.

(i) Which of the following angles does the dashed-ray direction make with the z-axis?

(ii) With the sample illuminated by *unpolarized* light, what is the *degree of polarization* for light scattered along the dashed-ray direction shown?

(iii) With the sample instead illuminated by *linearly polarized light* oriented along the y-direction (thus vertically polarized), which of the following statements is most accurate about the polarization of the light scattered along the x-axis and along the y-axis, respectively?

(iv) With the sample again illuminated by linearly polarized light oriented along the y-direction (thus vertically polarized), what is the polarization of the light emitted along the dashed-ray direction shown?

Solution 8.

(i) If we say the incident ray has vector direction $\mathbf{a}$, then $\mathbf{a} \cdot \hat{\mathbf{z}} = |\mathbf{a}| \cos\theta$ where θ is the angle we want. As in the hint, we can project down onto the xz plane, obtaining a vector with length $|\mathbf{a}| \cos\gamma_2$ and then project that vector down along z to obtain $\mathbf{a} \cdot \hat{\mathbf{z}} = |\mathbf{a}| \cos\gamma_2 \cos\gamma_1$ so

$$\theta = \arccos\left(\frac{|\mathbf{a}| \cos\gamma_2 \cos\gamma_1}{|\mathbf{a}|}\right) \approx 52.2°,$$

answer (B).

(ii)

$$P(\theta) = \frac{\sin^2\theta}{1 + \cos^2\theta} = 5/11 \approx 0.45,$$

answer (E).

(iii) The way I picture this is that the light excites a dipole oriented along y, which reemits vertically polarized light in the xz plane along its radiating toroid. This means that the emitted light is only along x and is vertically polarized, which corresponds to answer (C).

(iv) This time we are on the surface of the radiating toroid, and we've walked up it to an angle of γ_2, so the light, whilst still vertically polarized, is at an angle of 30° to the vertical (backwards, I want to say this is a negative angle), answer (A).